

Shawn Guo

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EDUCATION

Johns Hopkins University

BALTIMORE MD

B.S. in Computer Science; B.M. in Music for New Media; GPA 3.91/4.00

May 2026

- Relevant Courses: Computer Graphics and 3D Game Programming, Linear Algebra, Algorithms, Artificial Intelligence

EXPERIENCE

Software Engineering Intern, Audio

MIDDLETON WI

Activision/Raven Software

May 2025 - August 2025

- **Developed low-level C/C++ runtime systems and audio APIs** in a proprietary AAA engine for ***Call of Duty***
- **Extended a node-based, sample-aware character-dialogue system**, adding negative post-node delays, client-mask targeting, and visual debug tooling. Optimized dialogue systems to cut conversations memory use by approx. 40%
- Built **C#/.NET WPF audio tooling** for compression, legacy CSV migration, and VO-driven facial animation

Software Engineer, Audio

BALTIMORE MD

Snow Day Games LLC

March 2025 - Present

- Co-founded Snow Day Games and lead audio development for ***Vesuvius***, a \$15K grant-funded action-adventure
- Integrated **Wwise** with gameplay systems for ambience, combat, UI, and cutscenes in Unity
- Built a data-driven audio layer across 50+ event/state controls, **reducing runtime audio lookups by approx. 30%**

Systems and 3D Audio Programmer

REMOTE

Serious Point Games Ltd

March 2024 - May 2025

- **Designed and implemented a data-driven event architecture powering 90%+** of gameplay, UI, and audio interactions
- Unified Unity systems with **Wwise** runtime events, state changes, and designer-authored content hooks
- Built **Wwise Rooms & Portals spatial audio** workflows for room-aware reverb and dynamic propagation, reducing spatial-audio **memory overhead by approx. 30%** and lowering room reverb authoring time

Full Stack Engineer

BALTIMORE MD

Quest2Learn LLC

December 2023 - October 2024

- Built React Three Fiber learning modules and migration pipelines for a mobile/web AR education platform
- Developed reusable 3D UI, scene logic, and web interaction systems
- Migrated 40% of legacy Unity content into web-based workflows using JSON schemas

PROJECTS

Negative Space – Spatial Audio in a Custom C++ Engine | C++, raylib, DSP

December 2025 - Present

- **Built a custom C++ spatial audio stack** for the ***Negative Space engine*** to a full 10+ feature runtime audio layer, including 3D emitters/listeners, **partitioned convolution reverb**, attenuation, panning, and limiting
- Implemented **room-aware propagation**, obstruction/diffraction traces, and JSON-authored acoustic zones
- Optimized audio work out of the frame update path to **reduce frame drops by 75%+** compared to first iteration

Jam Audio Toolkit Package | C++, C#, Dear ImGui, miniaudio, Unity

March 2026 - June 2026

- Built a Unity audio implementation toolkit with reusable sound/music **ScriptableObject** types
- Added **runtime pooling**, custom inspectors, no-code triggers, filters, fades, and one-line APIs
- Created a **C++ Dear ImGui companion app** using **miniaudio** for live preview

TEACHING AND LEADERSHIP

Head TA | JHU **Game Engines Programming** (601.355) and **Computer Graphics & 3D Game Programming** (601.257)

President and Co-founder | **Hopkins Game Development Society**

SKILLS

Languages | C/C++, C#, Python, Java, Lua, TypeScript/JavaScript, HTML/CSS

Frameworks | React, Node.js, Tailwind, React Three Fiber, Unity, Unreal Engine 5

Tools & Middleware | Wwise, FMOD, Dear ImGui, .NET/WPF, Reaper, Pro Tools, Git, Perforce